

[DC] Virtual Reality for Mental Health: Toward Bioadaptive Narrative and Sensory Feedback Systems

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ABSTRACT

This doctoral thesis examines how virtual reality (VR), physiological biofeedback, and conversational AI can be combined to create adaptive mental health interventions. The project focuses on testing and developing systems that transition between narrative therapeutic dialogue and sensory-based grounding experiences based on real time indicators of user state. The aim is to advance accessible, non-pharmacological mental health support using consumer technologies and to understand when conversational versus sensory approaches are most effective, as well as explore the intersection and synergies between the two.

Index terms: Virtual Reality, mental health, biofeedback, physiological computing, conversational AI, adaptive systems, therapeutic readiness, Extended Reality.

1 INTRODUCTION

Mental health conditions remain a leading cause of disability worldwide, with access to effective care uneven and often insufficient [1]. Pharmacological treatments show limited efficacy for many forms of depression and related disorders [2], and although psychological interventions are effective, they are resource demanding and difficult to scale [3]. Digital and immersive technologies offer promising pathways to complement traditional care structures.

Biofeedback has long been used to support self-regulation and emotional stabilization by leveraging physiological signals such as HRV, EEG, and EDA [4,5]. VR environments strengthen attentional focus and therapeutic presence, with growing evidence supporting their usefulness across a range of mental health applications [6,7]. In parallel, conversational agents driven by large language models enable structured therapeutic exchanges, guided self-reflection, and scalable support [8,9].

Yet, these domains are usually explored independently. Systems rarely integrate physiological sensing with both immersive sensory modalities and narrative AI driven interaction. This project addresses that gap by proposing an adaptive VR system that can shift between sensory immersion and therapeutic dialogue based on physiological and linguistic indicators of the user's readiness for a given mode. The central aim is to understand how multimodal therapeutic systems can respond dynamically to user states in ways that enhance personalization, accessibility, and safety.

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2 APPROACH AND METHOD

The research follows an explorative and iterative design strategy aimed at developing and evaluating adaptive multimodal systems for consumer-grade VR headsets in combination with accessible physiological sensors capable of collecting EEG, HRV and EDA data. The overall approach integrates system development, physiological data acquisition and user centered evaluation to investigate how dynamic transitions between sensory immersion and narrative interaction can support therapeutic outcomes.

The framework is conceived around two complementary modes. The *sensory mode* emphasizes grounding and affective modulation through naturalistic VR environments, cyberdelic visuals, visual entrainment, spatial audio and haptic feedback. These experiences are designed to stabilize arousal and foster states that support further therapeutic engagement. The *narrative mode* centers on a generative AI agent offering guided conversation, reflective prompts and supportive dialogue that adapts to linguistic patterns and physiological indicators of user state.

Adaptive logic governs the transitions between these modes. Real time physiological markers, including HRV trends and EDA reactivity, are monitored to infer stress, attentional engagement and cognitive load. When users show signs of reduced readiness for conversational depth, the system prioritizes sensory grounding. When indicators suggest stability or openness, it transitions toward narrative exploration. This dynamic modulation aims to align the therapeutic style with moment-to-moment user needs.

Evaluation relies on a mixed methods design. Quantitative measures include HRV, EEG derived indices, EDA and validated psychological scales such as the Positive and Negative Affect Schedule (PANAS). Qualitative insights are gathered through semi-structured interviews and analyzed thematically to understand perceived safety, engagement and therapeutic value. Throughout the process, a human centered design perspective guides iterative refinement, ensuring that system behavior and interaction patterns remain responsive to user experiences and aligned with established principles in technology supported mental health care [10]. The following questions are of central importance to this research:

RQ1: How can physiological and linguistic indicators be combined to assess moment-to-moment therapeutic readiness in immersive environments?

RQ2: When does a narrative, AI-driven therapeutic dialogue offer the greatest benefit, and under what conditions is a sensory experiential mode more effective for supporting emotional regulation and engagement?

RQ3: What system architectures and design principles best support adaptive, multimodal VR mental health interventions using accessible consumer technology?

RQ4: How do users experience transitions between narrative and sensory therapeutic modes, and how do these transitions influence perceived safety, agency and therapeutic value?

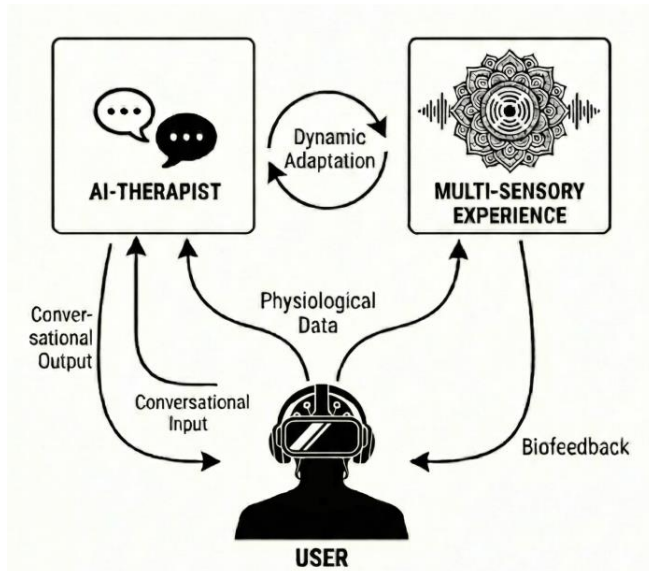


Figure 1: Dynamic adaptation between the narrative and sensory mode based on real-time physiological data from the user.

3 RESEARCH DIRECTION

The project proceeds in three successive yet interrelated phases. In the first phase, a comprehensive scoping review maps the current state of knowledge regarding AI-driven conversational agents in VR, biofeedback-informed immersive systems and hybrid interventions that combine narrative and sensory therapeutic modes. This establishes conceptual boundaries, identifies methodological gaps and informs system architecture. The second phase conducts empirical experiments in two parallel streams: one investigates narrative therapeutic dialogue delivered through conversational agents in VR augmented by physiological sensing, while the other explores purely sensory-based VR experiences designed to modulate arousal and support engagement. In the third phase these streams converge into an integrated adaptive framework: transitions between narrative and sensory modes are governed by real-time biomarkers and user behavioral/language cues to optimize therapeutic readiness. The framework articulates design principles for scalable, consumer-oriented interventions, and will be evaluated in mixed-method studies to assess system responsiveness, acceptability and initial therapeutic potential. To enhance clarity, the work could be divided into three main themes:

3.1 Theme One: AI-Driven Therapeutic Dialogue in Immersive XR Environments

This theme examines how conversational agents can deliver supportive, reflective and context-aware therapeutic dialogue within VR. It focuses on understanding how linguistic cues, user behaviour and physiological signals can shape the timing, depth and tone of AI-guided interactions. The work explores both opportunities and limitations of deploying narrative therapeutic processes in immersive settings.

3.2 Theme Two: Multisensory Therapeutic Experiences

This theme investigates how VR-based sensory environments, including naturalistic scenes, cyberdelic visuals, spatial audio and haptic feedback, can modulate arousal and emotional state. It

studies the mechanisms through which immersive sensory stimulation can support grounding, stress reduction and attentional shifts, and how these experiences prepare users for deeper therapeutic engagement.

3.3 Theme Three: Adaptive Integration Between Narrative and Sensory Modalities

This theme focuses on how real-time physiological and behavioural indicators can guide transitions between conversational and sensory modes. It aims to develop an adaptive framework that responds to moment-to-moment therapeutic readiness, determining when users benefit more from immersive grounding experiences and when narrative exploration is appropriate. The theme synthesizes insights from the first two themes into a unified, dynamically responsive system.

4 CONCLUSION

This research investigates how adaptive multimodal VR systems can integrate physiological biofeedback and conversational AI to support accessible and personalized mental health interventions. By examining real time transitions between sensory immersion and narrative engagement, the project aims to clarify how therapeutic readiness can guide moment to moment system behavior. The anticipated contributions include a conceptual framework for hybrid therapeutic interactions, a set of functioning prototypes that illustrate key system components, empirical insights into user experience and feasibility and design principles for consumer oriented digital mental health tools. Taken together, the work seeks to advance scalable, non-pharmacological approaches that respond dynamically to individual needs and broaden access to supportive psychological technologies.

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